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Quarto for reproducible research workflows and academic publishing

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Software

- Review
- Repository
- Archive

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Summary

Quarto is a multi-language, open-source scientific and technical publishing system that turns plain text and code into publication-ready papers, slides, websites, books, and teaching materials. It unifies writing and computation, supports the implementation of seamless reproducible workflows, and bridges the gap between conducting, writing, and communicating research (Wickham et al., 2023). As such, it is ideally suited for academic research and publishing.

The present article describes an accessible, hands-on introduction to Quarto for university students and academics with little to no prior experience with programming and markdown syntax. The online version of the tutorial is published on QuartoPub. It includes interactive quiz questions that encourage the reader to try things out immediately, providing immediate, motivating feedback. The tutorial itself was written in Quarto and the source code is published alongside the tutorial as an accompanying Open Educational Resource (OER) on Zenodo (Le Foll, 2026). As per its CC-BY license, it may be adapted, remixed, and reused in different educational contexts. Making use of Quarto's multi-format rendering options, a PDF version of the tutorial is also available for offline reading and annotation.

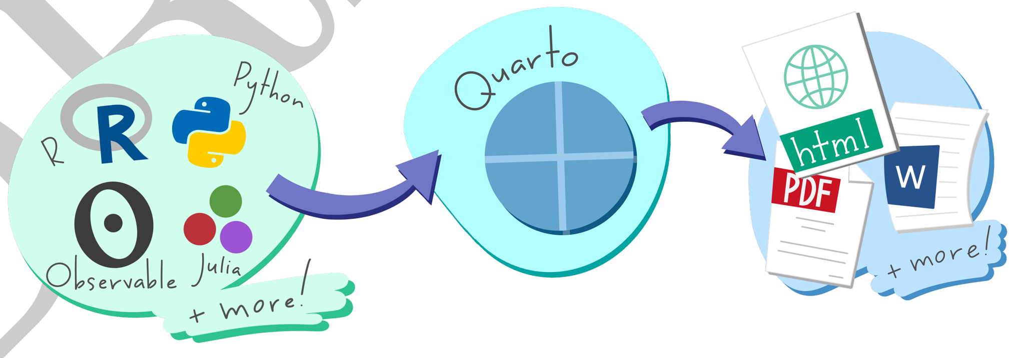

Figure 1: A schematic representation showing Quarto can understand multi-language input and produce multi-format outputs (Artwork CC BY 4.0 Allison Horst from the "Hello, Quarto" keynote by Julia Lowndes and Mine Çetinkaya-Rundel, first presented at the RStudio Conference 2022)

Statement of need

As a new system, accessible resources on and in Quarto are still relatively scarce. The official Quarto Guide provides extensive documentation; however, its "Get Started" pages assume prior knowledge that, in my experience, not all students and academics can be assumed to have, especially in the humanities and social sciences. Whilst the benefits of

Quarto for academic research have been highlighted from the launch of the new format in July 2022 (Allaire, 2022), early adopters have tended to come from disciplines where (statistical) programming has long become part and parcel of everyday research activities. Many of these colleagues were already using Rmarkdown and/or Jupyter Notebooks, from which switching costs to Quarto are low. In the humanities and social sciences, however, students and academics are still largely drafting their research outputs in (proprietary) text-processing software, foremost Microsoft Word, Google Docs, and Overleaf. For these students and colleagues, switching to Quarto represents a much steeper learning curve. As a result, this OER not only provides a hands-on introduction to Quarto, it also makes the case for how literate programming and multi-format exporting options have the potential to genuinely improve research workflows, by making them both more efficient and less error-prone. To this end, a step-by-step tutorial was designed that includes concrete examples and immediate opportunities for success.

The tutorial is an adaptation of Chapter 14 on "Reproducible research and academic writing in Quarto" in Le Foll (2025). The textbook chapter has been extensively tested and revised over multiple iterations following feedback from three cohorts of B.A., M.A., and Ph.D. students of linguistics, Romance languages, and language teaching at the University of Cologne, as well as students and colleagues from other institutions. The textbook, however, was written for linear reading and, as a result, is unsuitable as a standalone resource on Quarto for reproducible workflows and academic research. The tutorial described in the present article, by contrast, assumes no prior knowledge. It can be used both for self-study or as part of a course or workshop.

In contrast to most beginner-level Quarto tutorials, the present OER begins by outlining all installation steps in detail. It focuses on a single IDE, RStudio, that comes bundled with Quarto, in order to reduce cognitive load. It also relies on the use of locally-stored datasets, as opposed to data more conveniently accessible from an (R) package. The aim is to ensure that students and researchers who complete the tutorial have all the necessary keys in hand to be able to apply what they have learnt on their own research projects with their own data.

Educators are encouraged to adapt this OER to their students' needs and interests. The easiest way to achieve this is to change the dataset to something directly relevant to the target audience. The datasets currently used come from Dąbrowska (2019). They were chosen for three reasons. First, because they feature easily understandable variables that require little to no domain-specific knowledge. Second, because they are of interest to both linguistics and education students and researchers. And, last but not least, because they represent raw data featuring typical pitfalls of real-life research data such as inconsistent capitalisation, trailing whitespaces, and typos, that vividly illustrate the need for reproducible data wrangling workflows.

Learning objectives

As a beginner-friendly OER, "Quarto for reproducible research workflows and academic publishing: A step-by-step tutorial" aims to impart a basic understanding of literate programming and of the importance of reproducibility in research workflow.

The OER covers:

- how to install and use Quarto in RStudio
- how to write research reports, term papers, theses, and journal articles in Quarto by weaving together prose, code chunks, and code outputs
- how to format prose in markdown
- how to insert tables, images, plots with cross-references and alt-text in a Quarto document
- how to add and format bibliographic references in Quarto documents

- how to document package usage, as well as credit package developers with package references
- how to export and share Quarto documents in multiple formats including HTML, PDF, LibreOffice Writer, and Microsoft Word.

By the time learners have read the entire tutorial and completed all the tasks and quiz questions, they should have developed an understanding of the benefits of literate programming as implemented in Quarto, including how such a system can contribute to improving the reproducibility of their workflows. They are ready to start writing their own term paper, dissertation, thesis, research report, or journal article in Quarto and possess sufficient foundational knowledge with which they can start exploring some of the more advanced recommended resources listed at the end of the tutorial.

Instructional design

The tutorial includes clear step-by-step instructions illustrated with annotated screenshots with the aim of rendering maximally accessible to students and researchers from all disciplines, with a particular focus on humanities and social science scholars who may not have prior experience with literate programming, the use of IDEs such as RStudio, or YAML, Markdown, and BibTex syntax. To achieve this, the tutorial begins with an introduction to literate programming, explaining its benefits and focuses much of the writing process on using RStudio's visual mode for Quarto files, which is less intimidating than the source mode for learners with no programming experience. This approach helps to ease learners into the concept of weaving prose and code, showcasing how dynamic and reproducible research can be.

To make the learning experience engaging and relevant, the tutorial relies on real data from a published study (Dąbrowska, 2019). This not only serves to motivate learners, but also makes the newly acquired knowledge more readily implementable with their own data, given that the tutorial includes code to import the data into R. The HTML version of the tutorial includes mini-tasks that are based on the same study and aim to reinforce learning through hands-on practice. To encourage readers to explore Quarto during the completion of these tasks, twelve quiz questions with single or multiple-choice answers are integrated throughout the tutorial, providing immediate feedback to help learners assess their understanding. The conclusion section offers a tally of all correct responses, allowing learners to track their own progress without any penalties for multiple attempts. It also provides a list of learning objectives that they can tick off with cross-references to the corresponding sections of the tutorial, should they identify any gaps in their understanding at this stage.

The tutorial aims to introduce all relevant aspects of Quarto for academic writing and publishing in a single, cohesive document. Rather than overwhelming the reader with extensive details, the tutorial provides links to further resources throughout, as well as in the concluding "Going Further with Quarto" section. This approach ensures that learners have a solid foundation from which they can explore further resources detailing the endless more specialised options that Quarto has to offer.

Last but not least, the tutorial is designed with a strong emphasis on accessibility and inclusivity. The OER's Quarto source code is available for adaptation (including translation), allowing educators to tailor the content to specific learner groups. All images include captions and alt-text descriptions to support blind and low-vision readers. The HTML version also features a dark mode for enhanced readability. The PDF version ensures that the resource is also accessible in areas with unstable internet connections.

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Usage

'Quarto for reproducible research workflows and academic publishing: A step-by-step tutorial' has been tested in a variety of educational and professional contexts. It has been integrated as teaching material in semester-long courses on data analysis in R for linguistics and language education research at the University of Cologne. Furthermore, it has been used as standalone workshop material for doctoral and post-doctoral researchers from the humanities and social sciences. Additionally, it has served as a self-study learning and revision resource for students required to submit their term papers or dissertations in Quarto. Colleagues have also reported that it has been a valuable starting point for them when writing academic papers, research reports, or online supplements for the first time in Quarto.

Learner feedback has been overwhelmingly positive. Both students and seasoned researchers appear to appreciate the easy-to-follow instructions accompanied by screenshots, the concrete examples, and the mini-tasks and quiz questions that allow for immediate feedback. M.A. students and researchers were quick to grasp the benefits of literate programming in Quarto for research workflows and academic writing. Many expressed surprise at how accessible Quarto in RStudio is and were keen to continue exploring its capabilities. This positive reception underscores the tutorial's effectiveness in making Quarto approachable and genuinely valuable for a wide range of users within academia, including in traditionally less computational disciplines.

Acknowledgements

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